

MODESTO CITY, CA, USA

WMO: 724926

Lat: 37.624N

Lon: 120.951W

Elev: 73

StdP: 14.66

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 23258

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.2	33.7	21.0	15.8	55.6	24.7	18.9	54.4	22.5	53.7	19.6	52.3	2.3	120	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.2	101.7	69.4	98.6	68.3	95.8	67.2	71.6	96.3	70.0	94.3	68.7	92.3	9.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	84.5	82.6	60.5	78.8	79.6	58.8	74.3	77.6	35.3	96.4	33.9	94.4	32.8	92.4	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.0	15.4	DB	27.4	106.6	2.2	2.5	25.8	108.4	24.5	109.8	23.3	111.2	21.7	113.0	(4)
(5)			WB	26.6	73.8	2.3	2.0	25.0	75.2	23.6	76.4	22.3	77.6	20.7	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	47.7	51.8	56.9	61.3	68.5	75.3	79.6	78.1	74.0	64.9	54.6	47.7	(6)	
	DBStd	12.50	4.70	4.61	5.43	6.28	6.51	6.46	5.50	4.67	5.49	5.22	5.33	4.97	(7)	
	HDD50	261	100	31	7	2	0	0	0	0	0	0	17	104	(8)	
	HDD65	2260	535	368	255	144	36	5	0	0	3	64	314	536	(9)	
	CDD50	5166	29	83	220	340	574	760	918	871	721	462	154	34	(10)	
	CDD65	1691	0	0	4	32	145	314	453	406	275	61	1	0	(11)	
	CDH74	21782	0	1	113	544	1930	4102	5979	5041	3238	807	27	0	(12)	
	CDH80	11481	0	0	13	174	857	2236	3515	2819	1627	240	0	0	(13)	
(14) Wind	WSAvg	6.5	4.4	5.5	6.4	7.6	8.6	8.8	8.2	7.7	6.5	5.5	4.4	4.9	(14)	
(15) Precipitation	PrecAvg	12.4	2.2	1.9	2.1	1.0	0.2	0.1	0.0	0.1	0.3	0.7	2.0	1.7	(15)	
	PrecMax	27.4	5.7	5.9	6.5	3.0	1.1	0.5	0.7	0.7	2.2	2.4	6.0	3.9	(16)	
	PrecMin	6.4	0.2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	4.8	1.6	1.6	1.7	0.9	0.2	0.1	0.2	0.2	0.6	0.7	1.5	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.7	72.9	81.9	91.0	98.2	104.3	105.9	103.5	100.4	91.6	77.4	65.0	(19)	
		MCWB	53.4	58.0	60.8	64.6	67.6	70.5	71.2	69.2	67.5	64.1	60.5	55.2	(20)	
	2%	DB	62.6	69.0	77.1	84.8	93.6	100.0	102.0	99.6	96.6	85.9	73.4	61.6	(21)	
		MCWB	53.4	56.4	59.6	61.6	65.4	68.6	69.8	68.6	66.6	62.5	58.3	53.3	(22)	
	5%	DB	59.5	65.2	73.5	80.5	89.2	95.9	98.7	96.4	93.1	82.1	69.1	59.1	(23)	
		MCWB	52.1	54.4	58.0	59.8	63.2	66.9	68.8	67.6	65.9	60.8	56.7	52.5	(24)	
	10%	DB	56.7	62.1	69.4	75.7	84.4	91.9	95.5	93.3	89.6	78.6	65.2	56.8	(25)	
		MCWB	51.0	53.2	56.4	58.2	61.6	65.5	67.5	66.9	64.7	59.7	54.8	51.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.3	60.3	63.8	66.4	70.1	73.0	74.7	72.7	72.2	66.7	63.0	59.2	(27)	
		MCDB	62.1	67.8	77.0	86.0	94.5	100.4	100.5	98.1	91.7	84.6	73.7	62.6	(28)	
	2%	WB	55.2	58.1	61.3	63.5	67.0	70.3	71.9	70.5	69.0	64.5	60.3	56.3	(29)	
		MCDB	59.0	66.0	73.9	81.9	90.1	95.6	97.4	94.9	90.3	81.5	69.7	59.3	(30)	
	5%	WB	53.4	56.1	59.4	61.3	64.8	68.4	70.2	69.0	67.3	62.8	58.3	54.1	(31)	
		MCDB	57.3	63.4	70.9	77.2	86.1	92.4	94.9	92.9	89.1	78.2	66.2	57.2	(32)	
	10%	WB	51.6	54.2	57.3	59.3	62.6	66.7	68.6	67.7	65.8	61.1	56.5	52.0	(33)	
		MCDB	55.6	60.8	67.5	73.1	81.6	89.3	92.8	90.8	87.2	75.3	63.3	55.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	19.1	21.4	23.6	26.7	29.1	30.2	29.7	28.4	25.6	20.7	16.7	(35)	
		MCDBR	21.9	24.6	27.7	31.3	32.7	33.2	32.0	32.4	33.0	31.7	26.5	20.0	(36)	
	5% WB	MCWBR	13.3	13.8	12.9	12.5	11.7	11.4	11.3	11.3	10.7	12.1	13.2	12.2	(37)	
		MCWBR	15.8	21.5	25.3	28.6	30.9	31.6	30.3	30.5	30.2	27.5	22.2	12.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.339	0.335	0.341	0.358	0.362	0.355	0.366	0.350	0.352	0.353	0.343	0.326	(40)	
		taud	2.434	2.465	2.440	2.393	2.400	2.430	2.412	2.437	2.436	2.436	2.435	2.492	(41)	
	Ebn at Noon		266	284	293	292	291	292	288	291	285	273	262	264	(42)	
		Edh at Noon	27	30	33	37	37	36	37	35	33	30	27	24	(43)	
(44) All-Sky Solar Radiation	RadAvg		736	1065	1524	1969	2366	2646	2596	2344	1918	1382	912	667	(44)	
	RadStd		117	126	147	168	131	92	72	62	61	82	56	100	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.53	N/A	N/A	N/A	N/A	N/A	N/A	-256	+319	N/A	
(47)	Regional (47 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-216	+248	+58	

Nomenclature: See separate page